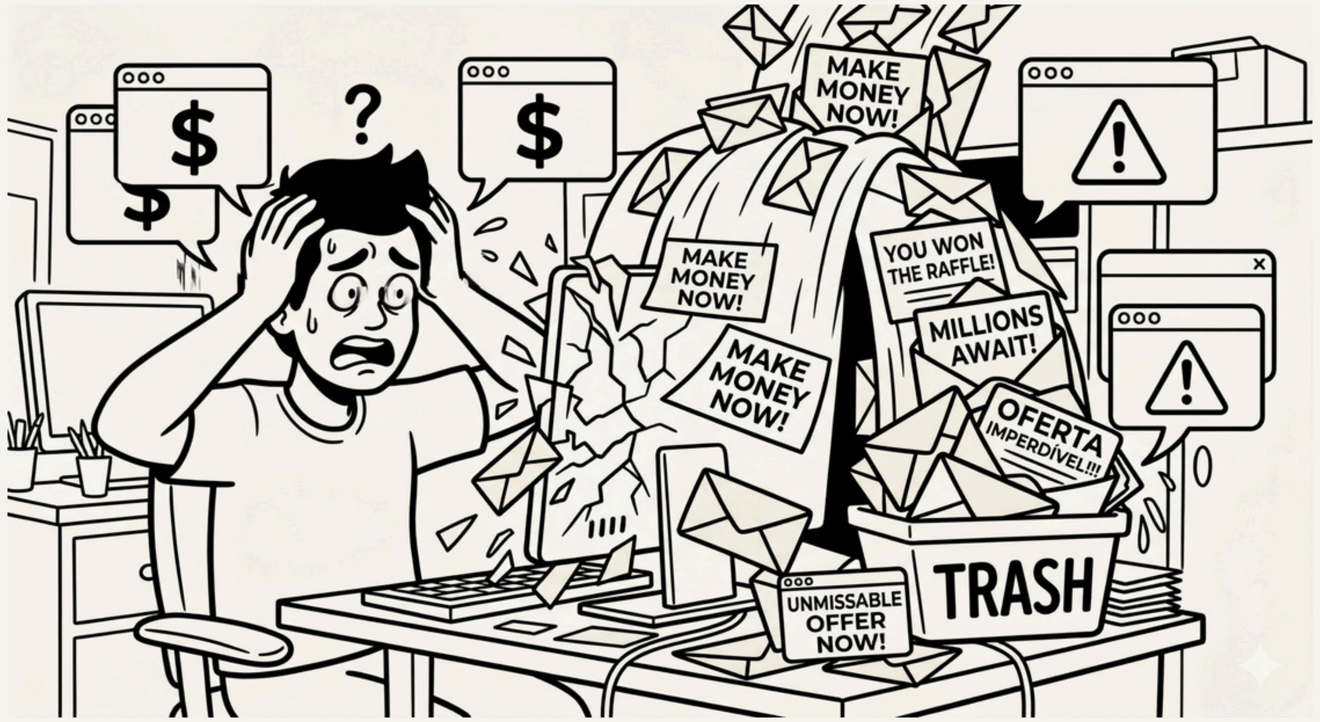
Email Classification

Ham/Spam/Phishing detection using Machine Learning Techniques



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Context of data

Phishing ≈ Ham

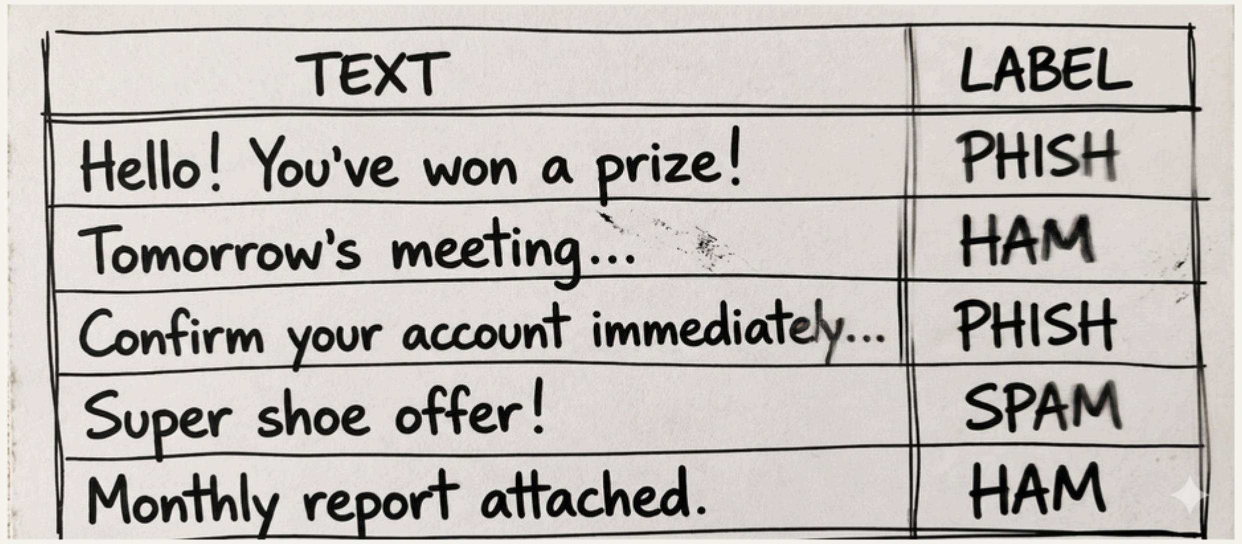
Phishing emails are designed to look like trusted messages to trick users

Data Provenance

- Dataset aggregated from multiple publicly available sources (Kaggle collection)
- Includes real-world data such as the **Enron** Email Corpus
- Combines corporate emails, spam, and phishing from different origins
- **Labels** are **inherited from original datasets** (not manually re-annotated)
- Spam and phishing samples come from automated sources (e.g., honeypots, benchmarks)

Dataset

- ~365k emails
- **Ham:** 45.3%; **Spam:** 39.3%; **Phishing:** 15.4%
- Language: English
- Large scale → enables robust models



TEXT	LABEL
Hello! You've won a prize!	PHISH
Tomorrow's meeting...	HAM
Confirm your account immediately...	PHISH
Super shoe offer!	SPAM
Monthly report attached.	HAM

Results Assignment 1

Model	Macro F1	Phishing Recall	Phishing Precision	Acc
MLP Optuna (TF-IDF)	96%	94%	92%	97%
MLP Weighted (Word2Vec) ★	96%	98%	90%	97%

★ *TF-IDF features treat words as isolated counts, missing semantic intent and context, which is exactly what phishing attackers exploit.*

DistilBERT + LoRA + cost-sensitive

Why DistilBERT?

40% smaller than BERT, 60% faster, retains ~97% language understanding
English general-domain pre-training (BookCorpus + Wikipedia) — same language as our data, but not the same genre → motivates domain adaptation later
Domain-specific alternatives (SecBERT, etc.) rejected: most are binary spam/ham and not reproducible at our scale

LoRA (PEFT):

- Freezes pre-trained weights, injects trainable low-rank matrices
- r=8, alpha=16, target modules: **q_lin**, **v_lin**
- Only 1.094% of parameters trained (740K out of 67.7M)

Cost-Sensitive Loss:

- Custom Trainer overrides CrossEntropyLoss with sklearn-balanced weights
- Weights: Ham 0.74 / Phishing 2.17 / Spam 0.85
- Forces larger gradient updates on missed phishing emails

Training Setup:

- Max 20 epochs, early stopping patience=3
- lr=2e-4, cosine schedule, fp16, batch size 32, AdamW
- ~5.5 hours on GPU

RoBERTa

Why RoBERTa?

Architectural sanity check: 125M parameters vs 66M (DistilBERT)
Removes Next Sentence Prediction, trained with larger batches and more data
Tests whether a larger model improves the performance ceiling

LoRA Setup:

- Same r=8, alpha=16, but target modules: query, value (RoBERTa naming)
- lora_dropout=0.05 (vs 0.1 for DistilBERT)

Cost-Sensitive Loss:

- Manually calibrated asymmetric weights: Ham 1.2 / Phishing 4.0 / Spam 0.3
- Stronger push on Phishing than DistilBERT's balanced weights
- **Rationale:** test whether a harder penalty on a larger architecture breaks the performance ceiling

Training Setup:

- Same pipeline as DistilBERT
- ~2-3 hours on GPU (faster convergence despite more parameters)

Domain Adaptation & Zero-Shot

Three transfer experiments:

MLM Continued Pre-training

- DistilBERT re-trained on raw unlabelled email corpus using Masked Language Modelling
- 264k texts, 3 epochs, 15% token masking probability
- Adapts internal representations to email vocabulary before classification fine-tuning
- **Validation perplexity:** 10.43 (generic DistilBERT typically 15–30 on specialised text)
- Then fine-tuned with exact same pipeline as Slide 4

Out-of-Domain Generalization (SMS)

- Fine-tuned model evaluated on SMS Spam Collection (5574 messages), never seen during training
- Tests whether learned cybersecurity semantics transfer across text domains
- No retraining, direct inference only

Zero-Shot Prompting (BART-MNLI)

- No fine-tuning, no labelled data
- Uses Natural Language Inference with descriptive class labels as hypotheses
- Evaluated on stratified 5000-email subset of the test set
- Baseline to quantify the value of fine-tuning in a security context

Experimental Results — Model Comparison

Model	Global Acc	Macro F1	Phishing Recall	Phishing Precision
MLP Optuna (TF-IDF) — A1	97%	96%	94%	92%
MLP Weighted (Word2Vec) — A1	97%	96%	98%	90%
DistilBERT Baseline	98.71%	98.72%	98.49%	99.14%
DistilBERT + LoRA + Weights ★	**98.76%**	**98.76%**	**98.43%**	**99.23%**
RoBERTa Cost-Sensitive	98.61%	98.65%	98.72%	98.98%
RoBERTa Baseline	98.63%	98.64%	98.25%	99.24%

★ DistilBERT + LoRA + Cost-Sensitive achieves +2.76pp Macro F1 over the best A1 model, with only 1.094% of parameters trained. Domain adaptation pushes it +0.08pp further.

Experimental Results

Model	Macro F1	Phishing Recall	Phishing Precision
DistilBERT + LoRA + Weights	98,76%	98,43%	99,23%
Domain-Adapted DistilBERT	98,84%	98,63%	99,22%
Delta	+0.08pp	+0.20pp	-0.01pp

Continued MLM pre-training on 264k unlabelled emails provides a consistent improvement at zero labelling cost.

Model	Macro F1	Phishing Recall	Phishing Precision
BART-MNLI (Zero-Shot)	35,50%	3,12%	12,77%
DistilBERT + LoRA + Weights	98,76%	98,43%	99,23%

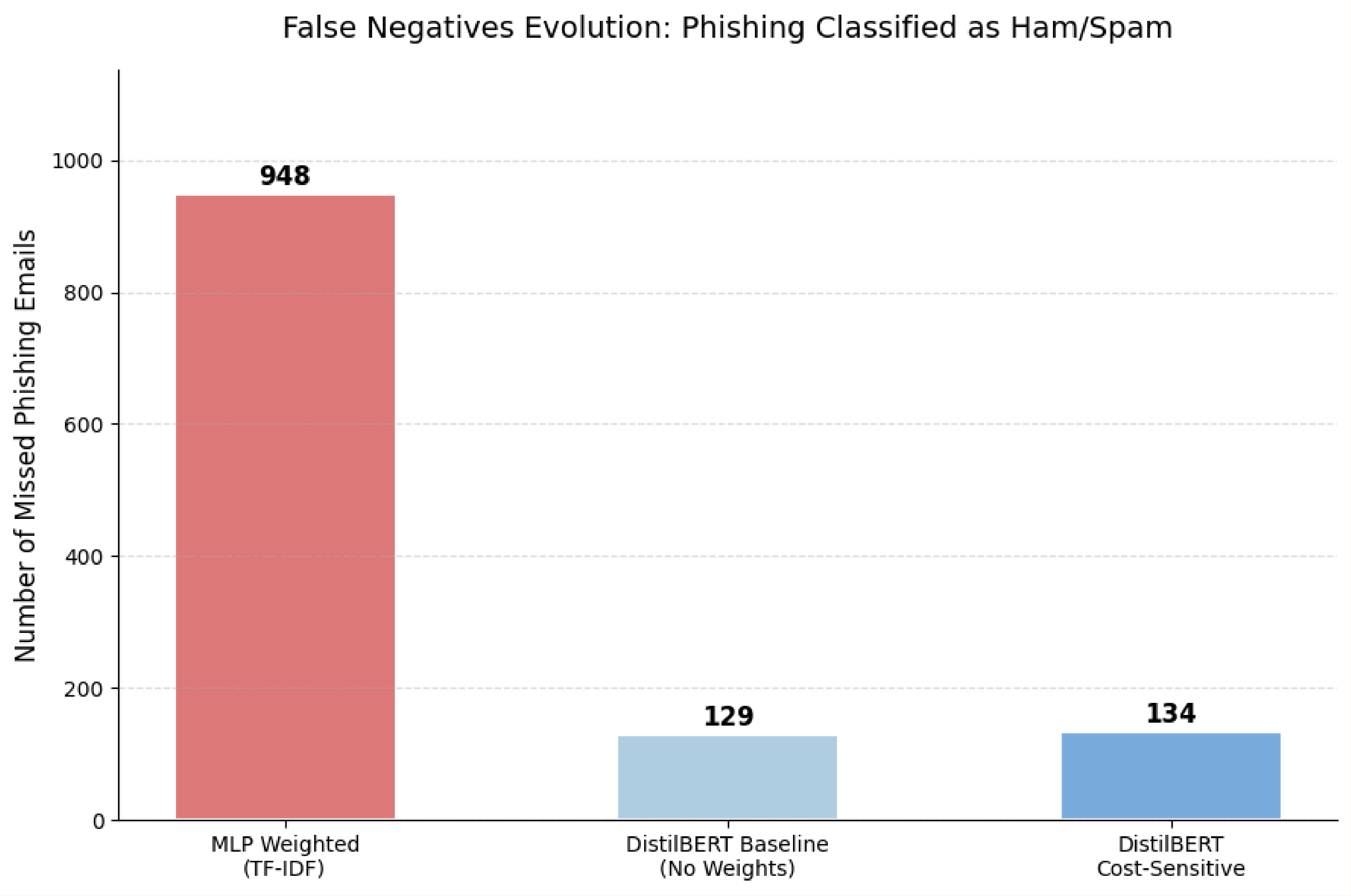
63pp gap in Macro F1 — zero-shot NLI cannot replace fine-tuning for security-critical tasks.

Despite never seeing SMS data during training, the model correctly flags 89% of SMS spam, showing that contextual embeddings capture transferable threat semantics. However, the 31% accuracy drop confirms that domain-specific fine-tuning remains essential for production deployment in new text media.

Metric	In-Domain (Email)	Out-of-Domain (SMS)
Accuracy	98,83%	68,00%
Malicious Recall	98,43%	89,00%

Fine-tuning with LoRA is essential. Zero-shot classification is not viable for production deployment in cybersecurity contexts.

Error Analysis — Persistent Blind Spots



Architecture change (TF-IDF → self-attention) reduced missed phishing by 820 emails.
Cost-sensitive weights barely moved the needle (129 vs 134, within run-to-run variance), the Transformer already saturates detection.

Persistent Blind Spot: "Word Salad" Evasion

- Semantic Collapse via "Word Salad"
- Random disconnected words dilute the semantic signal even for self-attention
- **Important Note:** all misclassified phishing went to Spam (2), never Ham (0) → a safer failure mode than Assignment 1, where missed phishing was delivered to the inbox (Ham).

Example blind spot (true Phishing → predicted Spam):

"best place buy viagra online best viagra price feeling better click away multitude books making us ignorant..."

Conclusions

Key Results

All four Transformer models converge to ~98.6–98.8% Macro F1, outperforming the best traditional MLP by ~2.7 pp. Domain-adapted DistilBERT achieves the highest Macro F1 (98.84%).

Recommended Model

DistilBERT + LoRA is the recommended production system: equivalent performance to RoBERTa with half the parameters (66M vs 125M), updating only 1.094% of weights via PEFT. ~134 undetected phishing emails across 55687 test emails.

Limitations

Adversarial drift from LLM-generated phishing; 256-token truncation loses content in long emails; subjective Spam/Phishing boundary introduces label noise; significant accuracy drop on out-of-domain data (SMS).

Future Work

- Domain adaptation pre-training (continued MLM on email corpus)
- Robustness evaluation against LLM-generated phishing
- Ensemble: DistilBERT embeddings + TF-IDF hybrid classifier
- Quantization / ONNX export for real-time inference

(Additional) State-of-the-Art Comparison

Model	Author	Class	Precision	Recall	F1-Score	Overall Accuracy	Macro F1
Naive Bayes	Ours (Baseline)	Ham	95%	95%	95%	90%	88%
		Phish	83%	75%	79%		
		Spam	87%	91%	89%		
	Alsuwit et al. (2024) [1]	Spam/Ham	97%	97%	97%	97%	
	Ghosh & Senthilrajan (2023) [2]	Spam/Ham	82%	85%	83%	87.63%	
	Manita et al. (2023) [3]	Spam/Ham	97%	95%	96%		
	Meléndez et al. (2024) [4]	Phish	98%	93%	96%	96%	
		Ham	95%	99%	97%		
Logistic Regression	Ours (Standard)	Ham	99%	99%	99%	96%	95%
		Phish	94%	89%	91%		
		Spam	96%	96%	96%		
	Ours (Weighted)	Ham	98%	97%	98%	95%	94%
		Phish	87%	94%	90%		
		Spam	95%	93%	94%		
	Alsuwit et al. (2024) [1]	Spam/Ham	98%	98%	98%	97%	
	Manita et al. (2023) [3]	Spam/Ham	73%	85%	78%		
	Meléndez et al. (2024) [4]	Phish	98%	99%	98%	98%	
		Ham	99%	98%	99%		
Random Forest (Standard)	Ours (Standard)	Ham	98%	99%	98%	96%	95%
		Phish	96%	86%	91%		
		Spam	93%	97%	95%		
	Alsuwit et al. (2024) [1]	Spam/Ham	98%	98%	98%	97%	
	Ghosh & Senthilrajan (2023) [2]	Spam/Ham	99%	99%	99%	99.91%	
	Fares et al. (2024) [5]	Phish	95%	92%	93%	95%	95%
		Ham	95%	92%	93%		
	Meléndez et al. (2024) [4]	Phish	98%	99%	99%	99%	
		Ham	98%	98%	98%		
Linear SVM	Ours (Standard)	Ham	98%	98%	98%	95%	94%
		Phish	94%	87%	90%		
		Spam	93%	96%	94%		
	Fares et al. (2024) [5]	Phish	94%	98%	96%	96.60%	97%
		Ham	99%	95%	97%		
	Meléndez et al. (2024) [4]	Phish	98%	99%	99%	99%	
MLP	Ours (Standard, TF-IDF)	Ham	99%	99%	99%	96%	95%
		Phish	93%	91%	92%		
		Spam	95%	96%	96%		
	Ours (Weighted, TF-IDF)	Ham	98%	99%	98%	94%	92%
		Phish	76%	98%	85%		
		Spam	98%	86%	92%		
	Ours (Optuna Tuned, TF-IDF)	Ham	99%	99%	99%	97%	96%
		Phish	92%	94%	93%		
		Spam	96%	96%	96%		
	Ours (Standard, Word2Vec)	Ham	99%	99%	99%	98%	97%
		Phish	97%	94%	96%		
		Spam	96%	98%	97%		
	Ours (Weighted, Word2Vec)	Ham	98%	99%	98%	97%	96%
		Phish	90%	98%	94%		
		Spam	98%	93%	96%		
Neural Network	Alsuwit et al. (2024) [1]	Spam/Ham	98%	98%	98%	98%	
BERT	Ghosh & Senthilrajan (2023) [2]	Spam/Ham	98%	99%	98%	98.67%	
CNN	Ghosh & Senthilrajan (2023) [2]	Spam/Ham	97%	98%	98%	98.50%	
roBERTa-base	Meléndez et al. (2024) [4]	Phish	99%	100%	100%	99%	
		Ham	100%	99%	99%		